Predicting UFC Fight Outcomes with Machine Learning

UE24CS352A Machine Learning | Mini-Project Report | October 2026

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1. Problem Statement

Given the pre-fight statistics of two UFC fighters (the Red corner and the Blue corner), predict who wins the bout. This is a binary classification problem: Class 1 means the Red fighter wins and Class 0 means the Blue fighter wins. We compare four algorithms (SGD classifier, MLP, Decision Tree, Gradient Boosting) and judge them against a trivial "always predict Red" baseline, using time-aware validation so that future fights never leak into training.

2. Dataset

We use a real UFC fight statistics CSV (`data/ufc_fights.csv`, the format recommended in the project README from the Kaggle UFC dataset). It has 6,012 fights and 144 columns, from 1994-03-11 to 2021-03-20. The target is the *Winner* column (Red, Blue or Draw).

Stage	Fights	Details
Raw data	6,012	1994 to 2021, includes 110 draws
Before 2010	1,244	1,229 Red wins, 0 Blue wins, 15 draws (98.8% Red)
Cleaned data	4,673	2010-01-02 to 2021-03-20, no draws; Red 2,750 (58.85%), Blue 1,923
Train set (older 80%)	3,738	2010-01-02 to 2019-04-13; Red win rate 58.64%
Test set (newer 20%)	935	2019-04-13 to 2021-03-20; Red win rate 59.68%

Data issue found and fixed. In fights before 2010 the dataset lists the winner as Red almost every time, which is a recording artefact rather than real fighting. It made the first time-series fold contain only one class, so cross-validation scores for SGD and Gradient Boosting came out as NaN, and it taught the models a false pattern. We drop all fights before 2010 (`MIN_YEAR = 2010`) and remove draws. After cleaning, the Red win rate is a realistic 58.85%.

3. Approach

- **Target and ordering:** Winner mapped to 1 (Red) or 0 (Blue); fights sorted by date.
- **Feature engineering:** each stat is the Red minus Blue difference, so it directly shows who has the edge (reach, age, height, weight, wins, losses, win streak, strikes landed and accuracy, takedowns, submission attempts, KO/TKO wins). 26 numeric difference features plus stance categoricals give 28 features. We also fixed a column-name mismatch (`win_by_KO/TKO`) that had silently dropped the KO/TKO feature.
- **Split and preprocessing:** chronological 80/20 split (no shuffling). Pipelines apply imputation, scaling and one-hot encoding.
- **Models:** SGD classifier (log loss, a linear baseline); MLP with hidden layers (64, 32), ReLU and early stopping; Decision Tree (max depth 6, min 10 samples per leaf); Gradient Boosting (100 estimators, learning rate 0.1, depth 4). No hyperparameter tuning was done.
- **Validation and metrics:** TimeSeriesSplit cross-validation (k=5) on the training data, and accuracy, precision, recall and F1 on the held-out test fights. The "Always Red" baseline is scored on the same test set.

4. Implementation Overview

File	Role
`src/preprocessing.py`	Loads and cleans data: removes draws and pre-2010 fights, parses dates, creates the target
`src/feature_engineering.py`	Builds the Red minus Blue difference features
`src/train_models.py`	Four scikit-learn pipelines and time-series cross-validation (checks each fold has both classes)
`src/evaluate_models.py`	Metrics, the Always Red baseline and feature importance
`train.py` / `app.py`	Training pipeline that saves models and results; Streamlit app where the user enters two fighters' stats and gets a predicted winner

5. Results

All scores are on the 935 held-out test fights (April 2019 to March 2021). CV is the mean and standard deviation over 5 time-series folds on the training data.

Model	Accuracy	F1 score	CV mean ± std
Always Red (baseline)	59.68%	44.61%	58.27%
SGD Classifier	54.44%	54.86%	55.54% ± 2.69%
MLP Classifier	**62.35%**	**60.57%**	**59.94% ± 2.39%**
Decision Tree	59.14%	59.14%	53.16% ± 2.26%
Gradient Boosting	59.89%	58.58%	58.72% ± 2.38%

Before cleaning, the same pipeline gave SGD 54.70%, MLP 60.03%, Decision Tree 55.72% and Gradient Boosting 60.54% accuracy against a baseline of 57.66%, with NaN cross-validation for SGD and Gradient Boosting. Cleaning fixed the NaN scores and raised the Decision Tree's test accuracy by 3.4 points.

6. Conclusions

- The MLP is the best model: 62.35% accuracy and 60.57% F1, which is 2.67 points above the baseline's accuracy and 15.96 points above its F1 (the baseline never predicts Blue, so its F1 is low).
- The gains are small. Gradient Boosting beats the baseline by only 0.21 points, and SGD and the Decision Tree score below it on accuracy. With 935 test fights the standard error of an accuracy score is about 1.6 points, so most gaps are close to noise. Fight outcomes are hard to predict from pre-fight statistics alone.
- The main lesson is about data quality: inspecting the label distribution revealed a recording bias that broke cross-validation, and comparing against a naive baseline kept the results honest.
- **Limitations and future work:** a single time split and no tuning; more data and features (age at fight time, days since last fight, weight-class models); XGBoost, Random Forest and ensembles; GridSearchCV or Optuna; class weighting; live data from UFCStats. This project is for learning and not for betting.

Team contribution: [Name 1]: [what they did]. [Name 2]: [what they did].